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Excellent

Addis Ababa Science and Technology University

Mathematics for Natural Science [Math 1007]

Date: June 9, 2022 Time allowed: 60 min Maximum Score 15 % **Test-I**

Name [redacted] **Id No.** [redacted] **Section** [redacted]

Part I: Write "True" if the statement is correct and "False" if the statement is Wrong on the space provide (5 points)

True 1. If the truth values of p, q and r are T, F and F, respectively, then the truth value of $(\neg p \wedge q) \vee r$ is F.

True 2. If $(p \wedge q) \Leftrightarrow (p \vee q)$ and $\neg p$ are $true, then$ q is $false$

False 3. If $\neg p \Rightarrow q$ is $False, then$ $\neg(q \wedge \neg p)$ is $false$

True 4. For two sets A and B, $A \triangle B = B \triangle A$.

True 5. If $A \subseteq B^c$, then $A \cap B = \phi$.

Part II: Short Answer Questions (4 points)

Write the Most simplified answer in the space provided

1. The truth value of $(\exists y \in \mathfrak{R})(\forall x \in \mathfrak{R})(x + y^2 \geq y)$ is True.
2. If $A_n = \{x\colon \forall x \in N, (x = n) \Rightarrow (x < n)\}$, then $\cap_{i=1}^{n} A_i =$ $\{x : x > n\}$
3. Let A={ ϕ, { ϕ }} and B= { ϕ ,{ ϕ ,{ ϕ }}}, then B \ A = $\{\{\phi, \{\phi\}\}\}$
4. The number of proper subset of the set S={a, b, c,5} is $2^n - 1 = 2^4 - 1 = 16 - 1 = 15$, $n = 4$
5. If the truth value of $(\neg p \vee q) \wedge (p \Leftrightarrow q)$ is false, then the truth value of $\neg p \Rightarrow q$ is True

p-T
q-F

Part IV: Workout Questions

1. Using truth table ,determine the validity of $\neg p \Rightarrow \neg q, q \vdash p$ [2 pts]

p	q	$\neg p$	$\neg q$	$\neg p \Rightarrow \neg q$
T	T	F	F	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

Valid

If you see 1^{st} row, all premises are true & conclusion aswell.

b/c of 1^{st} row

2. In a survey of 250 families, it was found that 220 listen radio in the morning while 125 listen during the evening. If 120 families listen radio during the morning and evening, how many families listen radio: [3 pts]

(a) at least once a day?
(b) Only once a day?
(c) Neither in the morning nor in the evening

(a) $n(M \cup E) = n(M) + n(E) + n(M \cap E)$
$= 100 + 5 + 120$
$= 225$

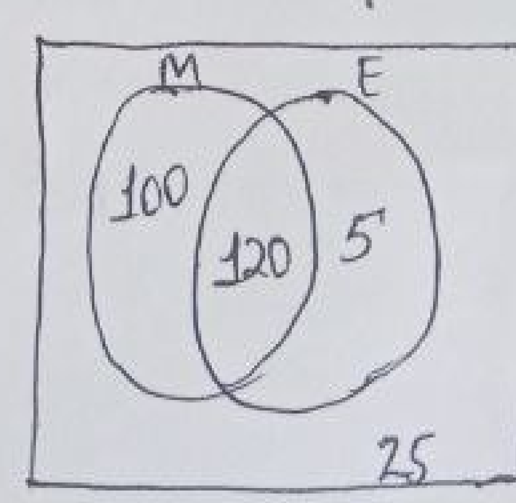
or ~~n(M)+n(E) = 220+125~~
$= \cancel{5}$

$n(M \cup E) = n(M) + n(E) - n(M \cap E)$
$= 220 + 125 - 120$
$= 225$

(b) $\cancel{M' \cap E'} = 100 + 5 = 105$
$M \setminus E \cup E \setminus M = M \Delta E$

(c) $(M \cup E)' = U - (M \cup E)$
$= 250 - 225$
$= 25$

M - morning Radio listener
E - evening "



$U = 250$
$n(M) = 220$
$n(E) = 125$
$n(M \cap E) = 120$

$M \setminus E = 220 - 12$
$= 100$
$E \setminus M = 125 - 12$
$= 5$